

NETWORKS AND COMPLEXITY

Solution 19-10

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 19.10: Matrix perturbation theory [Lvl. 4]

Let's suppose we make a small change to our Jacobian such that $\mathbf{J} = \mathbf{A} + p\mathbf{B}$, where $\mathbf{A}$ is the original Jacobian, $\mathbf{B}$ describes the change and p is a parameter that is initially zero. Let's assume we know the eigenvalues λ_n and the left eigenvectors $\mathbf{w}_n^\dagger$ and the right eigenvectors $\mathbf{v}_n$ of $\mathbf{J}$ for $p = 0$. We now compute how the eigenvalues change as we increase p :

- a) Start with the (right) eigenvector equation for $\mathbf{J}$, substitute $\mathbf{J}$ and differentiate the whole equation with respect to p . Finally eliminate $\mathbf{A}$ by using the substitution backwards.

Solution

We start with

$$\mathbf{J}\mathbf{v}_n = \lambda_n\mathbf{v}_n \quad (1)$$

Substituting

$$\mathbf{J} = \mathbf{A} + p\mathbf{B} \quad (2)$$

yields

$$(\mathbf{A} + p\mathbf{B})\mathbf{v}_n = \lambda_n\mathbf{v}_n \quad (3)$$

When we change p both the eigenvectors and eigenvalues will change so we need to use the product rule in the differentiation

$$\partial_p(\mathbf{A} + p\mathbf{B})\mathbf{v}_n = \partial_p(\lambda_n\mathbf{v}_n) \quad (4)$$

$$\mathbf{B}\mathbf{v}_n + \mathbf{J}\mathbf{v}_n' = \lambda_n'\mathbf{v}_n + \lambda_n\mathbf{v}_n' \quad (5)$$

In the second row of this equation I used a dash to indicate the derivative with respect to p . I have also already put the $\mathbf{J}$ back in by undoing the substitution in the second row.

- b) You should now have one equation that contains the change of $\partial_p\lambda_n$ but also $\partial_p\mathbf{v}_n$, which we don't want to compute. To fix this, multiply your equation with $\mathbf{w}_n^\dagger$ from the left, then find a way to eliminate $\partial_p\mathbf{v}_n$, and solve for $\partial_p\lambda_n$.

Solution

Multiplying $\mathbf{w}_n^\dagger$ from the left yields

$$\mathbf{w}_n^\dagger\mathbf{B}\mathbf{v}_n + \mathbf{w}_n^\dagger\mathbf{J}\mathbf{v}_n' = \mathbf{w}_n^\dagger\lambda_n'\mathbf{v}_n + \mathbf{w}_n^\dagger\lambda_n\mathbf{v}_n' \quad (6)$$

Our goal is to remove the $\mathbf{v}_n'$ on both side of the equation. This makes you wonder if they maybe cancel each other, doesn't it? Recall that

$$\mathbf{w}_n^\dagger\mathbf{J} = \lambda_n\mathbf{w}_n^\dagger. \quad (7)$$

Substituting this relationship yields

$$\mathbf{w}_n^\dagger\mathbf{B}\mathbf{v}_n + \mathbf{w}_n^\dagger\lambda_n\mathbf{v}_n' = \mathbf{w}_n^\dagger\lambda_n'\mathbf{v}_n + \mathbf{w}_n^\dagger\lambda_n\mathbf{v}_n' \quad (8)$$

Now the terms are indeed identical and we can cancel them which leads to

$$\mathbf{w}_n^\dagger \mathbf{B} \mathbf{v}_n = \mathbf{w}_n^\dagger \lambda'_n \mathbf{v}_n \quad (9)$$

Since λ'_n is a number we can just pull it out from between the two vectors and solve for

$$\frac{\partial \lambda_n}{\partial p} = \lambda'_n = \frac{\mathbf{w}_n^\dagger \mathbf{B} \mathbf{v}_n}{\mathbf{w}_n^\dagger \mathbf{v}_n}. \quad (10)$$

- c) Does the result give you an idea for another way in which we could rank the importance of nodes?

Solution

The equation gives us a way to estimate the effect on the eigenvalues if we change the matrix. Suppose I wanted to improve the stability of the system and I had a way of adding a small controller to one node, which provides a little bit of negative feedback without changing the steady state. In that case I might want to rank the nodes in terms of their effect on the eigenvalue.

If I chose to control node i , the matrix $\mathbf{B}$ has only one non-zero element $B_{ii} = -1$, and hence we can compute the impact on the rightmost eigenvalue as

$$\frac{\partial \lambda_1}{\partial p} = -\frac{w_{1,i}^\dagger v_{1,i}}{\mathbf{w}_1^\dagger \mathbf{v}_1}. \quad (11)$$

which acts as a kind of ‘rightmost eigenvector perturbation centrality’.